Pranav Balaji Pooruli

9701 AS Organic Chemistry Reactions

Alkane + oxygen

- Type of reaction: **Combustion**

Complete combustion

`alkane + oxygen -> carbon dioxide + water`

Incomplete combustion

`alkane + oxygen -> carbon monoxide + water`

Alkane + halogen

- Type of reaction: **Substitution**
- Mechanism: **Free-radical substitution (FRS)**

`alkane + halogen -> halogenoalkane + hydrogen halide`

- Ultraviolet light (usually sunlight)
  - Provides activation energy
  - Induces homolytic fission

Specifics

- Initiation step
  - Homolytic fission of X-X
    - e.g. Cl-Cl
  - Gives 2X·
    - e.g. Cl-Cl -> 2Cl·
- Propagation steps
  - alkane + X· -> alkyl· + HX
    - e.g. CH_4 + Cl· -> CH_3· + HCl
- Termination steps
  - alkyl· + X· -> halogenoalkane
    - e.g. CH_3· + Cl· -> CH_3Cl
  - alkyl· + alkyl· -> alkane
    - e.g. CH_3· + CH_3· -> C_2H_6

Alkene + hydrogen

- Type of reaction: **Addition (hydrogenation)**
- Mechanism: **Electrophilic addition**

`alkene + hydrogen -> alkane`

- Finely divided platinum/nickel catalyst
- Heat (200℃)
- Pressure (700 kPa)

Alkene + hydrogen halide

- Type of reaction: **Addition**
- Mechanism: **Electrophilic addition**

`alkene + hydrogen halide -> halogenoalkane`

- Room temperature

Specifics:

- With longer, asymmetric alkanes, there are always two possible products
- For major product:
    - Hydrogen goes to the carbon with
        - More H
        - Less alkyl
    - Halogen goes to the carbon with
        - Less H
        - More alkyl

Alkene + steam

- Type of reaction: **Catalytic addition (hydration)**
- Mechanism: **Electrophilic addition**

`alkene + steam` ⇌ alcohol

- Concentrated phosphoric(V) acid
- High temperature
- High pressure

Specifics

- Cannot result in a tertiary alcohol
    - Unless a hydrogen on a carbon with a double bond is substituted for an alkyl group

Alkene + halogen

- Type of reaction: **Addition**
- Mechanism: **Electrophilic addition**

`alkene + halogen -> dihalogenoalkane`

- Room temperature
- Halogen can be in aqueous solution or in standard state

Specifics:

- Carbon-carbon double/triple bonds decolorize bromine water, i.e. aqueous bromine - $Br_2(aq)$

Alkene + [O]

- Type of reaction: **Oxidation**

With cold dilute acidified manganate(VII)

`alkene + water + [O] -> diol`

- Cold dilute acidifed potassium manganate(VII)
- Cold meaning room temperature

Specifics:

- Each carbon which is part of the carbon-carbon double bond receives a hydroxy group
- A way to test for unsaturation (bromine water test is more common)

With hot concentrated acidified manganate(VII)

- The portion of the alkene on either side of the double bond oxidizes separately

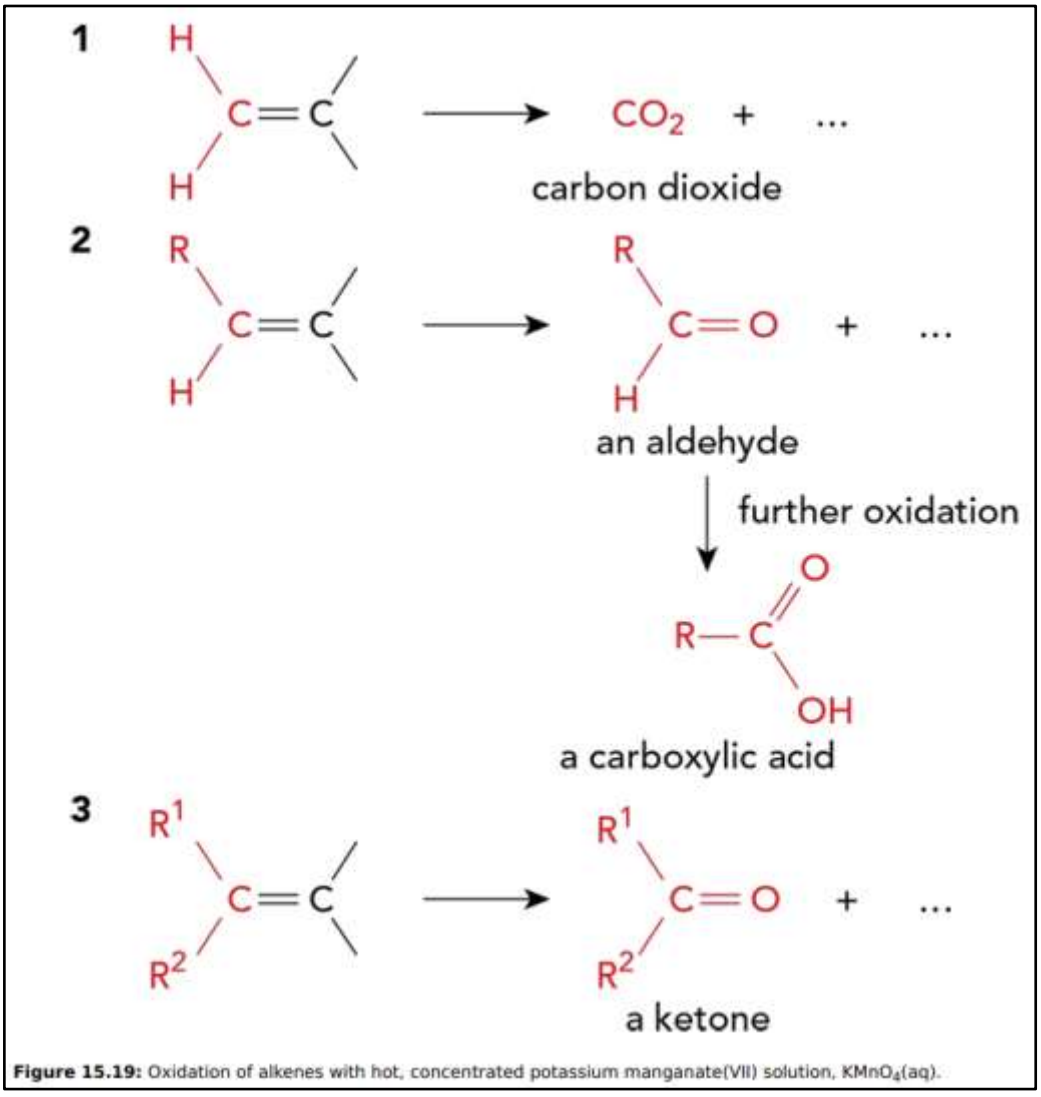


Figure 15.19: Oxidation of alkenes with hot, concentrated potassium manganate(VII) solution, $KMnO_4(aq)$.

- When CO_2 is produced, H_2O is also produced
    - Because the hydrogens from the alkene portion are removed
- Known as oxidative cleavage

Halogenoalkane + hydroxide ion

Aqueous hydroxide ions

- Type of reaction: **Substitution**
- Mechanism: **Nucleophilic substitution**

`halogenoalkane + hydroxide ion -> alcohol + halide ion`

- Aqueous hydroxide ions
- Heat
- Reflux

Specifics

- Hydroxide ion acts as a nucleophile, donating electron pair to the carbon which is bonded to the halogen

Ethanolic hydroxide ions

- Type of reaction: **Elimination**

`halogenoalkane + hydroxide ion -> alkene + water + halide ion`

- Ethanolic hydroxide ions
- Heat
- Reflux

Hydrolysis of halogenoalkanes

- Type of reaction: **Hydrolysis**
- Mechanism: **Nucleophilic substitution**

`halogenoalkane + water -> alcohol + proton + halide ion`

- Heat
- Reflux

Specifics:

- Similar to reaction with aqueous alkali
- Difference is that this reaction is slower
    - O in H-OH has δ^- charge
    - Whereas O in hydroxide has full negative charge
    - So O from H-OH is a less effective nucleophile
- Fastest nucleophilic substitution reactions happen with iodoalkanes, slowest with fluoroalkanes
    - C-I bond is longest and weakest, lowest bond energy, easy to break
    - C-F bond is shortest and strongest, highest bond energy, difficult to break

Halogenoalkane + cyanide ion

- Type of reaction: **Substitution**
- Mechanism: **Nucleophilic substitution**

`halogenoalkane + cyanide ion -> nitrile + halide ion`

- Ethanolic potassium cyanide
- Heat
- Reflux

Specifics:

- Cyanide ion adds a carbon to the carbon chain
- Useful reaction for extending the carbon chain
- Example
  `bromoethane + cyanide ion -> propanenitrile + bromide ion`

Halogenoalkane + ammonia

- Type of reaction: **Substitution**
- Mechanism: **Nucleophilic substitution**

`halogenoalkane + ammonia -> amine + hydrogen halide`

- Excess ethanolic ammonia
- Heat under pressure

Specifics:

- When a primary amine is formed
  e.g. `bromoethane + ammonia -> ethylamine + hydrogen bromide`
  (ethylamine is a primary amine)
  - If ammonia is *not* in excess, primary amine behaves as a nucleophile
  - Primary amine attacks halogenoalkane
  - Forms secondary amine
    `bromoethane + ethylamine -> diethylamine + hydrogen bromide`
    $C_2H_5Br + C_2H_5NH_2 \rightarrow (C_2H_5)_2NH + HBr$
- Actual final product is an ammonium halide, because the hydrogen halide formed is acidic and is immediately neutralized by the alkaline ammonia
  e.g. $HBr + NH_3 \rightarrow NH_4Br$

Mechanisms of nucleophilic substitution with halogenoalkanes

Primary halogenoalkanes - S_N2

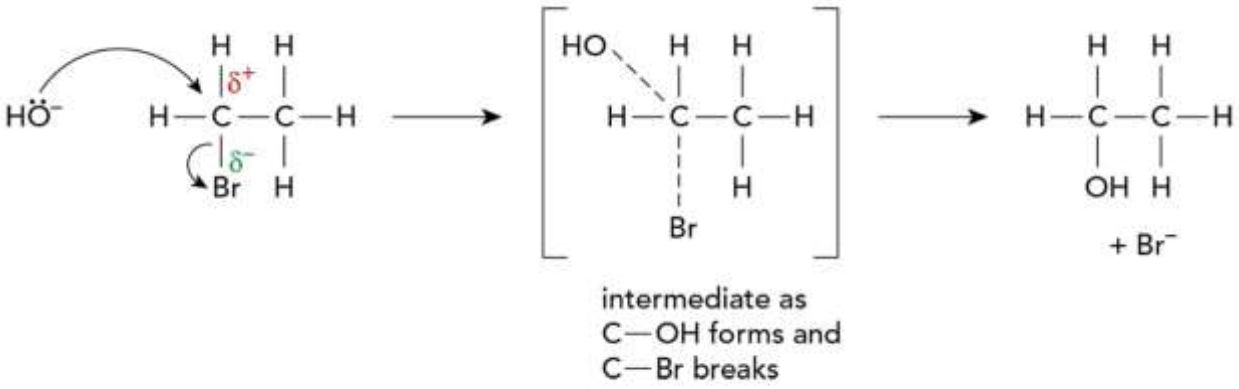


Figure 16.5: The mechanism of nucleophilic substitution in a primary halogenoalkane.

- C-X bond breaks heterolytically
- C-OH bond forms simultaneously
- S_N2
  - S - substitution
  - N - nucleophilic
  - 2 - indicates that the RoR depends on the concentration of 2 reacting species (both halogenoalkane and hydroxide ions in this case)
- OH^- nucleophile attacks carbon atom with δ^+ charge
- First step in mechanism involves both reacting species
- Stereospecific - Nucleophile attacks carbon on opposite side of leaving group

Tertiary halogenoalkanes - S_N1

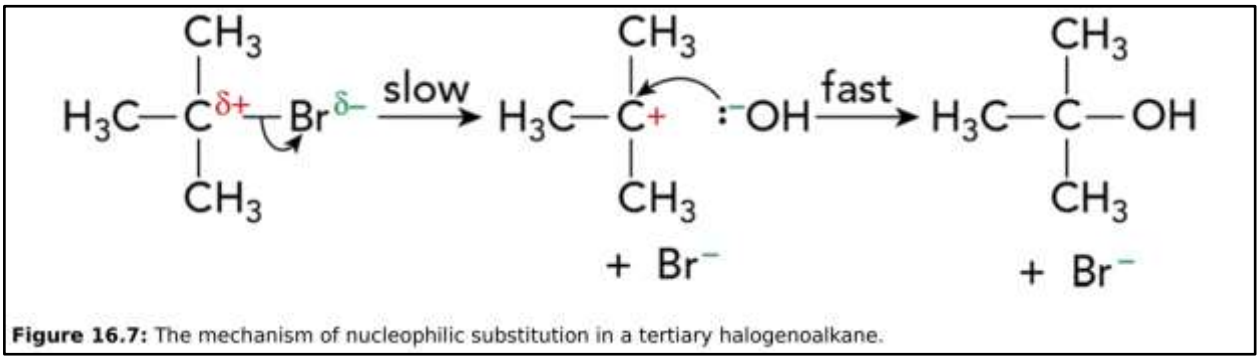


Figure 16.7: The mechanism of nucleophilic substitution in a tertiary halogenoalkane.

- S_N1 because RoR only depends on the concentration of 1 reagent, the halogenoalkane in this case
  - For both S_N2 and S_N1, the RoR is determined by the slow step in the mechanism
- C-X bond breaks heterolytically
- OH^- nucleophile attacks carbocation
- First step in mechanism involves only the halogenoalkane (heterolytic fission of C-X bond)

Secondary halogenoalkanes

Nucleophilic substitution in a secondary halogenoalkane uses a mixture of the S_N1 and S_N2 mechanisms

Alcohol + water

- Partial dissociation of alcohols in water

$alcohol \rightleftharpoons$ alkoxide ion + proton

- All species in aqueous solution

Specifics:

- Position of equilibrium lies in the left
- Concentration of H^+ is lower than in $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$
    - Alkyl group in alkoxide ion has positive inductive effect
    - This concentrates the negative charge on the oxygen atom
    - So the alkoxide ion accepts H^+ more readily than the OH^- ion
- Alcohols are weaker acids than water
- E.g. $CH_3OH(aq) \rightleftharpoons CH_3O^-(aq) + H^+(aq)$

Alcohol + oxygen

- Type of reaction: **Combustion**

Complete combustion

```
alcohol + oxygen -> carbon dioxide + water
```

Specifics:

- Ethanol burns with a clean blue flame in a good supply of air

Incomplete combustion

```
alcohol + oxygen -> carbon monoxide + water
```

Alcohol + halide ions

- Type of reaction: **Substitution**
- Mechanism: **Nucleophilic substitution**

With hydrogen halide

```
alcohol + hydrogen halide -> halogenoalkane + water
```

- Dry hydrogen halide gas
- Room temperature

Specifics:

- To make HCl:
  $NaCl(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HCl(g)$
  - Concentrated sulfuric acid
- To make HBr:
  $KBr(s) + H_3PO_4(l) \rightarrow KH_2PO_4(s) + HBr(g)$
  (or)
  $2KBr(s) + H_3PO_4(l) \rightarrow K_2HPO_4(s) + 2HBr(g)$
  (or)
  $KBr(s) + H_2SO_4(l) \rightarrow KHSO_4(s) + HBr(g)$
  - Concentrated phosphoric(V)/sulfuric acid
- Steps
  - Hydroxy group accepts proton from acid
  - C-O bond becomes easier to break
  - Nucleophilic substitution by halide ion
  - Water given off

With sulfur dichloride oxide

alcohol + sulfur dichloride oxide -> halogenoalkane + hydrogen chloride + sulfur dioxide

- Room temperature

Specifics:

- Sulfur dichloride oxide - $SOCl_2$
- HCl and SO_2 evolve as gas
- E.g. $CH_3OH + SOCl_2 \rightarrow CH_3Cl + HCl + SO_2$

With phosphorus(V) chloride

alcohol + phosphorus(V) chloride -> halogenoalkane + hydrogen chloride + phosphorus(V) oxychloride

- Room temperature

Specifics:

- Phosphorus(V) chloride - PCl_5
- Phosphorus(V) oxychloride - $POCl_3$
- Phosphorus(V) chloride must be in solid state
- Release of $HCl(g)$ makes steamy white fumes
  - Used as a test for the hydroxy group
- E.g. $CH_3OH + PCl_5 \rightarrow CH_3Cl + HCl + POCl_3$

With phosphorus(III) halide

alcohol + phosphorus(III) halide -> halogenoalkane + phosphorous acid

- Heat

Specifics:

- For bromo- or iodoalkanes, warm the alcohol with red phosphorus + bromine/iodine to make PBr_3 or PI_3
- E.g. $3CH_3OH + PI_3 \rightarrow 3CH_3I + H_3PO_3$

Alcohol + Group I metal

`alcohol + metal -> metal alkoxide + hydrogen`

Specifics:

- `O-H` bond breaks
    - As opposed to `alcohol + hydrogen halide` where `C-O` bond breaks as `-OH` group is lost
- Similar to `water + metal`, but less vigorous
- The longer the hydrocarbon chain in the alcohol, the less vigorous the reaction with a given Group I metal
    - Positive inductive effect pushes electron density towards O
    - `O-H` bond is stabilized, and more difficult to break
- For example:
  `propan-1-ol + sodium -> sodium propoxide + hydrogen`
  $2C_3H_7OH + Na \rightarrow 2C_3H_7O^-Na^+ + H_2$

Alcohol + carboxylic acid

- Type of reaction: **Condensation (esterification)**

`carboxylic acid + alcohol` ⇌ ester + water

- Heat
- Reflux
- Concentrated phosphoric(V)/sulfuric acid
    - Dehydrating agent

Hydrolysis of esters

Acid hydrolysis

`ester + water` ⇌ carboxylic acid + alcohol

- Dilute sulfuric acid
- Heat
- Reflux

Specifics:

- Reversible reaction, both sides will always be present

Alkaline hydrolysis

`ester + hydroxide ion -> carboxylate ion + alcohol`

- Dilute aqueous alkali
- Heat
- Reflux

Specifics:

- Not reversible, so all ester broken down with excess alkali

Dehydration of alcohols

- Type of reaction: **Elimination (dehydration)**

`alcohol -> alkene + water`

- Alcohol in vaporized state
- Heat
- Catalyst - any of these
  - Hot aluminum oxide powder
  - Pieces of porous pot or pumice
  - Concentrated acid

Specifics:

- Gaseous alkene collected in inverted measuring cylinder in trough of water

Alcohol + [O]

Primary alcohol

`alcohol + [O] -> aldehyde + water`

- Acidified aqueous potassium dichromate(VI)
- Warm
- Distill to prevent further oxidation (to carboxylic acid)
- Further oxidation:
  `aldehyde + [O] -> carboxylic acid`
  - Excess acidified potassium dichromate(VI)
  - Heat
  - Reflux

  Specifics:
  - This reaction also works with `HCOOH`, methanoic acid, because it also has an aldehyde functional group, `-CHO`
  - The opposite (reduction of the aldehyde group in methanoic acid) does not work

Secondary alcohol

`alcohol + [O] -> ketone + water`

- Acidified aqueous potassium dichromate(VI)
- Warm
- No distillation needed, as ketone cannot be oxidized further

Tertiary alcohol

- Does not oxidize because there are no hydrogens that can be removed from the carbon atom which is bonded to the hydroxy group
- When warmed with acidified potassium dichromate(VI), the solution stays orange, whereas for primary/secondary alcohols, it turns green from chromium(III) ions

Carboxylic acid + water

- Partial dissociation of carboxylic acids in water

`carboxylic acid` ⇌ carboxylate ion + proton

- All species in aqueous solution

Specifics:

- Weak acid due to partial dissociation
- E.g. `HCOOH(aq)` ⇌ $HCOO^-(aq) + H^+(aq)$

Carboxylic acids take part in typical acid reactions

Carboxylic acid + [H]

- Type of reaction: **Reduction**

```
carboxylic acid + 4[H] -> alcohol + water
```

- Lithium tetrahydridoaluminate dissolved in dry ether
- Room temperature

Specifics:

- Lithium tetrahydridoaluminate - $LiAlH_4$
- Liquid ether is dried because $LiAlH_4$ reacts violently with water
- Sodium tetrahydridoborate, $NaBH_4$, is too weak to reduce a carboxylic acid
- This reaction always forms a primary alcohol
- E.g. `HCOOH + 4[H] -> CH3OH + H2O`

Aldehyde/ketone + [H]

- Type of reaction: **Reduction**

```
aldehyde + 2[H] -> primary alcohol
ketone + 2[H] -> secondary alcohol
```

With sodium tetrahydridoborate

- Warm the aldehyde/ketone…
- …with an aqueous alkaline solution of sodium tetrahydridoborate, $NaBH_4$

With lithium tetrahydridoaluminate

- Add lithium tetrahydridoaluminate, $LiAlH_4$, dissolved in a dry ether (e.g. diethyl ether)
- At room temperature
- Organic solvent must be dry because $LiAlH_4$ is more powerful than $NaBH_4$ and will react vigorously with water

Aldehyde/ketone + hydrogen cyanide

- Type of reaction: **Addition**
- Mechanism: **Nucleophilic addition**

`aldehyde/ketone + hydrogen cyanide -> 2-hydroxynitrile`

- Heat
- Aqueous solution
- Mixture of KCN + HCN is used
  - KCN is the source of CN^-
  - HCN is the source of H^+
  - HCN is a weak acid: $HCN \rightleftharpoons H^+ + CN^-$
    - KCN being a strong base reduces $[H^+]$, shifting the equilibrium to the right
    - KCN helps HCN dissociate

Specifics:

- Useful reaction as it extends the carbon chain
- For example
  - `butanal + hydrogen cyanide -> 2-hydroxypentanenitrile`
    $C_3H_7CHO + HCN \rightarrow C_3H_7CH(OH)CN$
  - `butanone + hydrogen cyanide -> 2-hydroxy-2-methylbutanenitrile`
    $C_2H_5COCH_3 + HCN \rightarrow C_2H_5C(OH)(CN)CH_3$

Hydrolysis of nitriles

Acid hydrolysis

`nitrile + hydrochloric acid + water -> carboxylic acid + ammonium chloride`
$-C{\equiv}N + HCl + 2H_2O \rightarrow -C(=O)OH + NH_4Cl$

- Dilute hydrochloric acid
- Heat
- Reflux

Alkaline hydrolysis

nitrile + hydroxide ion + water -> carboxylate ion + ammonia

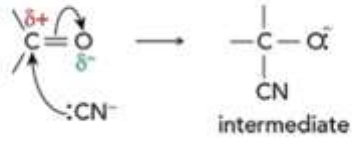
$-C{\equiv}N + OH^- + H_2O \rightarrow -C(=O)O^- + NH_3$

- Dilute aqueous sodium hydroxide
- Heat
- Reflux

Mechanism of nucleophilic addition in aldehydes/ketones

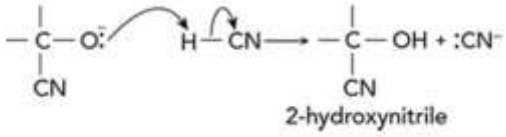
- Carbonyl group, >C=O, is polarized due to high electronegativity of O
- Bonding electrons are attracted by O, giving C a δ^+ charge
- Carbonyl C atom becomes open to attack by a nucleophile, like the CN^- ion
- In CN^-, the negative charge is on the C atom, not the N atom

Step 1:



- This intermediate is highly reactive and quickly reacts with a proton
- This H^+ may be from HCN, from dilute acid, or from water

Step 2:

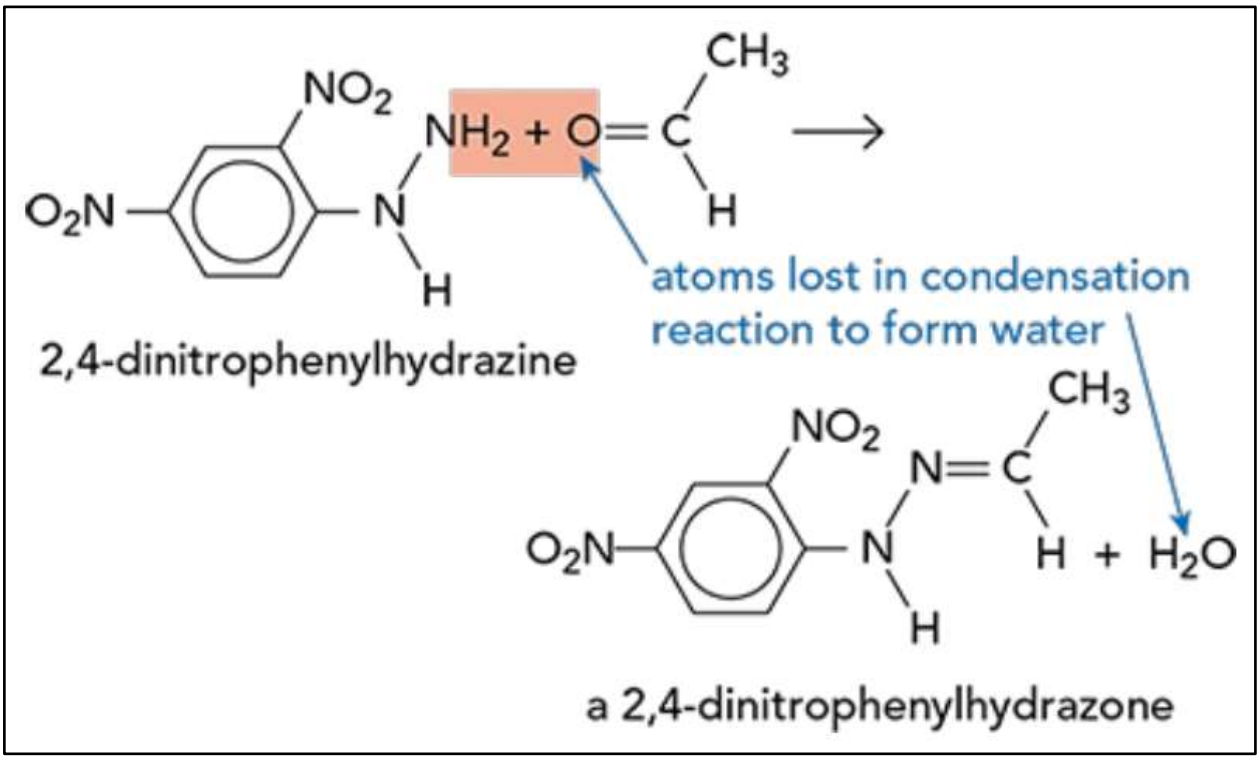


- All aldehydes and ketones form '2-hydroxynitriles' when they undergo nucleophilic addition with HCN
    - Because the hydroxy group is always on the carbon atom adjacent to the nitrile group
    - Exception is methanal, HCHO, which forms $CH_2(OH)CN$ which is simply hydroxyethanenitrile

Testing for aldehydes/ketones

2,4-DNPH test

- Test for **carbonyl >C=O group in aldehydes and ketones**
- Type of reaction: **Condensation**



Reaction of ethanal with 2,4-DNPH

Specifics:

- 2,4-DNPH - 2,4-dinitrophenylhydrazine
- 2,4-DNPH is added as aqueous solution in a test tube
- Reacts with aldehydes and ketones
    - No reaction with carboxylic acids and esters…
    - …despite the presence of a >C=O group
- Formation of a **deep-orange precipitate** indicates a positive result
    - The precipitate is a 2,4-dinitrophenylhydraz**o**ne
- Identity of original carbonyl comopund is found by
    - Recrystallizing the 2,4-dinitrophenylhydrazone precipitate
    - Measuring its melting point
    - Referring to melting point data

Tollens' reagent

- Test for **distinguishing between aldehydes and ketones**
- Type of reaction: **Redox**



Figure 18.4: The 'before' and 'after' observations when Tollens' reagent is warmed with an aldehyde, such as ethanal, CH_3CHO.

Specifics:

- Tollens' reagent - ammoniacal silver nitrate solution
    - $AgNO_3(aq)$ in excess $NH_3(aq)$
- Carbonyl compound is warmed with colorless Tollens' reagent in a test tube
- Reacts with aldehydes
    - Ag^+ ions act as a mild oxidizing agent
    - $RCHO$ oxidized to $RCOOH$ and Ag^+ reduced to Ag
        - Selective oxidation (hydroxy groups are not oxidized)
    - $RCOOH$ immediately neutralized to $RCOO^-$
    - Does not react with ketones or carboxylic acids…
        - …as they cannot be further oxidized
- Formation of a **silver mirror** indicates a positive result
    - Silver mirror forms from silver atoms collecting on the inside of the test tube
    - Called 'silver mirror test'

Fehling's solution

- Test for **distinguishing between aldehydes and ketones**
- Type of reaction: **Redox**



Figure 18.5: 'The 'before' and 'after' observations when Fehling's solution is warmed with an aldehyde, such as ethanal, CH_3CHO.

Specifics:

- Fehling's solution - alkaline solution containing copper(II) ions
    - Contains $Cu^{2+}(aq)$ and $OH^-(aq)$
- Carbonyl compound is warmed with blue Fehling's solution in a test tube
- Reacts with aldehydes
    - Cu^{2+} ions act as an oxidizing agent
    - $RCHO$ oxidized to $RCOOH$ and Cu^{2+} reduced to Cu^+
        - Selective oxidation (hydroxy groups are not oxidized)
    - $RCOOH$ immediately deprotonated by OH^- to form H_2O
    - Does not react with ketones or carboxylic acids…
        - …as they cannot be further oxidized
- Formation of an **opaque red/orange precipitate** indicates a positive result
    - The precipitate is copper(I) oxide, Cu_2O

Tri-iodomethane test

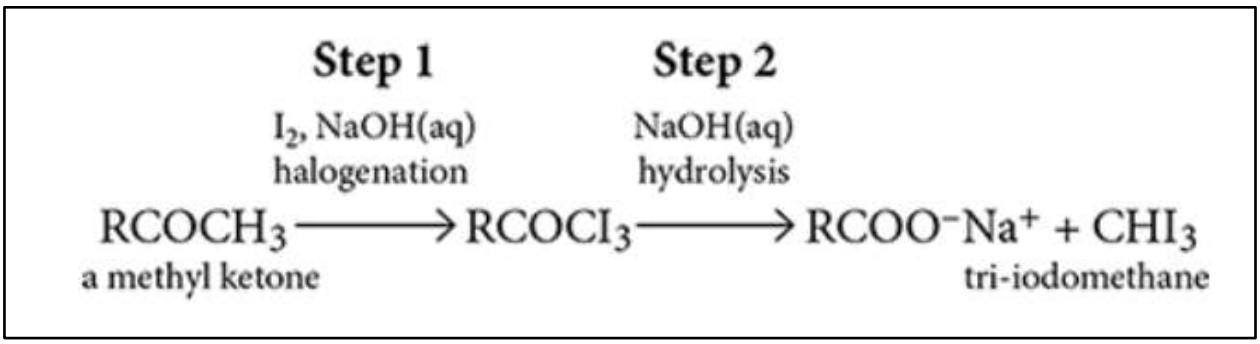
- Test for **$CH_3C(=O)$- group (methyl ketones and ethanal)**
- Type of reaction: **Halogenation + hydrolysis**



Figure 18.6: The yellow precipitate of tri-iodomethane forming.

Specifics:

- Tri-iodomethane (aka iodoform) - CHI_3
- Carbonyl compound is warmed with an alkaline solution of iodine in a test tube
    - Solution contains $I_2(aq) + OH^-(aq)$
- Reacts with $RC(=O)CH_3$ in two steps
    1. Halogenation - 3H replaced by 3I
       $RC(=O)CH_3 \rightarrow RC(=O)CI_3$
        - $I_2(aq)$ and $NaOH(aq)$
    2. Alkaline hydrolysis - forms $RCOO^-(aq) + CHI_3(s)$
       $RC(=O)CI_3 \rightarrow RC(=O)O^- + CHI_3$
        - $NaOH(aq)$



- Can also react with $CH_3CH(OH)$ -
    - $CH_3CH(OH)$ - first oxidized to $CH_3C(=O)$ - by alkaline $I_2(aq)$
- In all, gives positive result with
    - Methyl ketones
    - Ethanal

    - $CH_3CH(OH)$- type alcohols
- Formation of a **yellow precipitate** indicates a positive result
    - The precipitate is $CHI_3(s)$
    - The other product is $RCOO^-Na^+$, a sodium carboxylate salt

Reactions that require **reflux**

- Halogenoalkane + hydroxide ion
- Hydrolysis of halogenoalkanes
- Halogenoalkane + cyanide ion
- Esterification
- Hydrolysis of esters
- Oxidation of aldehydes (with excess acidifed $K_2Cr_2O_7(aq)$)
- Hydrolysis of nitriles

Reactions that are **reversible**

- Hydration of alkenes
- Dissociation of alcohols in aqueous solution
- Esterification
- Acid hydrolysis of esters
- Dissociation of carboxylic acids in aqueous solution

Pranav Balaji Pooruli

9701 AS Inorganic Chemistry Reactions

Period 3 element + chlorine

- All reactions require heating

Sodium

$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

Specifics:

- Vigorous reaction

Magnesium and aluminum

$Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$
$2Al(s) + 3Cl_2(g) \rightarrow Al_2Cl_6(s)$

Specifics:

- Vigorous reaction
- $MgCl_2(s) + NaOH(aq)$ forms a white precipitate of $Mg(OH)_2(s)$ which is insoluble in excess
- $AlCl_3(s) + NaOH(aq)$ forms a white precipitate of hydrated $Al(OH)_3(s)$ which dissolves in excess to form $NaAl(OH)_4(aq)$

Silicon

$Si(s) + 2Cl_2(g) \rightarrow SiCl_4(l)$

Specifics:

- Slow reaction

Phosphorus

$P_4(s) + 10Cl_2(g) \rightarrow 4PCl_5(s)$

- Excess chlorine gas

Specifics:

- PCl_5 - phosphorus(V) chloride
- Slow reaction

Period 3 metal + water

Sodium

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

- Works even with cold water

Specifics:
- Vigorous reaction
- Sodium metal melts and moves around surface of water
- Molten ball gets smaller and smaller until it disappears
- Strongly alkaline solution formed

Magnesium

Cold water

$Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(aq) + H_2(g)$

Specifics:
- Very slow reaction
- $Mg(OH)_2$ is only slightly soluble…
- …so weakly alkaline solution formed
- pH ≈ 9.5

Steam

$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$

Specifics:
- Vigorous reaction

Aluminum oxide + alkali

$Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2NaAl(OH)_4(aq)$
- Room temperature

Specifics:
- $NaAl(OH)_4$ - sodium tetrahydroxoaluminate

Silicon dioxide + alkali

$SiO_2(s) + 2NaOH(aq) \rightarrow Na_2SiO_3(aq) + H_2O(l)$
- Hot concentrated alkali

Specifics:
- Na_2SiO_3 - sodium silicate

Phosphorus(V) oxide + water

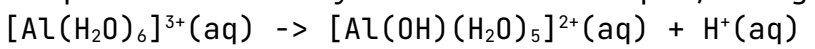
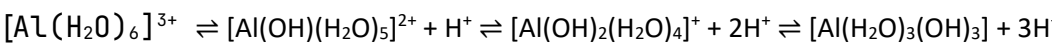
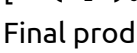
$P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$

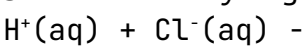
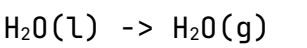
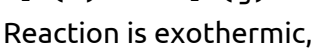
Specifics:
- Vigorous reaction

Hydrolysis of period 3 chlorides

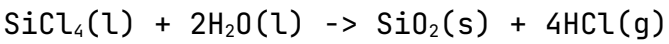
Aluminum

1 Hydration and dissociation into a complex hexaaquaaluminum(III) ion and a chloride ion


2 Deprotonation of the hydrated aluminum complex, making the solution <u>acidic</u>


Final product of hydrated  is insoluble, but is high enough the keep the position of equilibrium to the left, so <u>no solid is observed</u>

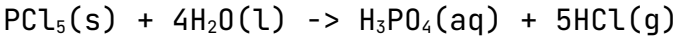
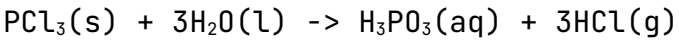
3 Liberation of hydrogen chloride gas and steam


Reaction is exothermic, so water evaporates and  forms, observed as <u>misty white fumes</u>

Silicon



Specifics:
-  forms off-white precipitate
-  dissolves to give acidic solution
- Dense white fumes of  seen

Phosphorus

Specifics:
- Vigorous reaction
- Both products highly soluble…
- …highly acidic solution formed
- Dense white fumes of seen

Halide ions + conc. H_2SO_4

- Type of reaction: **Redox**

Chloride ions

$NaCl(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HCl(g)$

Specifics:

- Acid added to sodium chloride crystals in small drops
- $HCl(g)$ is denser than air

Bromide ions

$NaBr(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HBr(g)$

Specifics:

- Oxidation of $HBr(g)$ / reduction of $H_2SO_4(l)$ follows
  $2HBr(g) + H_2SO_4(l) \rightarrow Br_2(g) + SO_2(g) + 2H_2O(l)$
  - Reddish brown bromine gas seen

Iodide ions

$NaI(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HI(g)$

Specifics:

- Oxidation of $HI(g)$ / reduction of $H_2SO_4(l)$ follows
  $2HI(g) + H_2SO_4(l) \rightarrow I_2(g) + \underline{SO_2(g)} + 2H_2O(l)$
  $6HI(g) + H_2SO_4(l) \rightarrow 3I_2(g) + \underline{S(s)} + 4H_2O(l)$
  $8HI(g) + H_2SO_4(l) \rightarrow 4I_2(g) + \underline{H_2S(g)} + 4H_2O(l)$
  - H_2SO_4 reduced to 3 different sulfur products
    - Sulfur dioxide
    - Sulfur
    - Hydrogen sulfide
  - Sulfur seen as yellow solid
  - Hydrogen sulfide gas has strong smell of rotten eggs
  - Iodine is produced as a purple vapor

Chlorine + alkali

- Type of reaction: **Disproportionation**

Cold alkali

$Cl_2(aq) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$

- Cold alkali - 15°C

Specifics:

- $NaClO$ - sodium chlorate(I)
- Ionic equation
  $Cl_2(aq) + 2OH^-(aq) \rightarrow Cl^-(aq) + ClO^-(aq) + H_2O(l)$

- Ionic half-equations of chlorine
    - 0 to -1
      $\frac{1}{2}Cl_2 + e^- \rightarrow Cl^-$
    - 0 to +1
      $\frac{1}{2}Cl_2 + 2OH^- \rightarrow ClO^- + H_2O + e^-$

Hot alkali

$3Cl_2(aq) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$

- Hot alkali - 70°C

Specifics:

- $NaClO_3$ - sodium chlorate(V)
- Ionic equation
  $3Cl_2(aq) + 6OH^-(aq) \rightarrow 5Cl^-(aq) + ClO_3^-(aq) + 3H_2O(l)$
- Ionic half-equations of chlorine
    - 0 to -1
      $\frac{1}{2}Cl_2 + e^- \rightarrow Cl^-$
    - 0 to +5
      $\frac{1}{2}Cl_2 + 6OH^- \rightarrow ClO_3^- + 3H_2O + e^-$

Chlorine + water

$Cl_2(aq) + H_2O(l) \rightarrow HCl(aq) + HClO(aq)$

Specifics:

- $HClO$ - chloric(I) acid, aka hypochlorous acid
- Chloric(I) acid partially dissociates to form chlorate(I) ions
- Chloric(I) acid and chlorate(I) ions sterilize water by killing bacteria

Oxidation of nitrogen

Step 1

$N_2(g) + O_2(g) \rightarrow 2NO(g)$

Specifics:

- NO - nitrogen(II) oxide
- High activation energy provided by lightning

Step 2

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

Specifics:

- NO_2 - nitrogen(IV) oxide

- NO oxidized by atmospheric oxygen

Step 3

$2NO_2(g) + H_2O(l) + ½O_2(g) \rightarrow 2HNO_3(aq)$

Specifics:
- Nitrogen(IV) oxide dissolves in water droplets in the air, plus oxygen
- Dilute nitric acid falls to Earth in rain
- Provides NO_3^- source for plants
- Contributes to acid rain

Ammonium salt + base

$NH_4^+(s) + OH^-(s) \rightarrow NH_3(g) + H_2O(l)$
- Heat
- Solid salt

Specifics:
- Acid-base reaction
  - NH_4^+ acts as Brønsted-Lowry acid
  - OH^- acts as Brønsted-Lowry base
- Example
  $2NH_4Cl(s) + Ca(OH)_2(s) \rightarrow CaCl_2(s) + 2H_2O(l) + 2NH_3(g)$

Formation of acid rain

Formation of gases

- Source of $N_2(g)$ - Atmospheric nitrogen
- Source of $SO_2(g)$ - Combustion of fossil fuels

1. $N_2(g) + O_2(g) \rightarrow 2NO(g)$
   a. Happens under high temperature AND high pressure
2. $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
3. $SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$
4. $NO_2(g) + ½O_2(g) \rightarrow NO_2(g)$ (NO_2 regenerated)

Oxides of nitrogen are only produced under the
- high temperature
- high pressure

inside an internal combustion engine when atmospheric $N_2(g)$ and $O_2(g)$ come into contact

- Nitrogen(IV) oxide catalyzes the oxidation of sulfur dioxide (sulfur(IV) oxide)…
- …to sulfur trioxide (sulfur(VI) oxide)

Formation of acids

1. $2NO_2(g) + H_2O(l) + ½O_2(g) \rightarrow 2HNO_3(aq)$
2. $2NO_2(g) + H_2O(l) \rightarrow HNO_2 + HNO_3(aq)$
3. $SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$

- These acids fall to Earth in acid rain

Inside a catalytic converter

$2CO(g) + 2NO(g) \rightarrow N_2(g) + 2CO_2(g)$

- Hot catalyst of platinum/rhodium/palladium

Flame colors

Period 3

Na	yellow
Mg	white
Al	
Si	none
P	white
S	blue
Cl	none

Group 2

Be	none
Mg	white
Ca	red
Sr	red
Ba	green

Group 1

Li	red
Na	yellow
K	lilac